

1. Administrative & Study Identification

Full Study Title: EPIK-P4: A Phase II Single-arm Study to Assess the Efficacy, Safety and Pharmacokinetics of Alpelisib (BYL719) in Pediatric and Adult Patients With PIK3CA-related Overgrowth Spectrum (PROS) **Short Title/Acronym:** EPIK-P4 **Protocol Number:** CBYL719F12202 **NCT Number:** NCT06997588 **Sponsor Name:** Novartis **Investigational Product:** Alpelisib (BYL719 / Vijoice / Piqray) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the efficacy of Alpelisib (ALP) based on the proportion of participants achieving a confirmed objective response by a Blinded Independent Review Committee (BIRC). A confirmed objective response requires a radiological response, confirmed by a subsequent assessment performed at least after 4 weeks, demonstrating a $\geq 20\%$ reduction from baseline in the sum of target lesion volumes. **Secondary Objectives:** To evaluate the efficacy, safety, pharmacokinetics (PK), and tolerability of ALP in adult and pediatric participants with PROS by assessing changes in target lesion volumes, non-target lesion volumes, and all MRI-measurable lesion volumes.

2.2 Background & Rationale PROS is a group of rare clinical syndromes caused by tissue-specific somatic mutations in the *PIK3CA* gene, leading to the activation of the AKT and mTOR signaling pathways and resulting in progressive tissue overgrowth. Currently, there is no cure, and standard treatments involve invasive procedures. Alpelisib, a small-molecule PI3K α -specific inhibitor, has shown efficacy in severe PROS. The EPIK-P4 study is designed to evaluate the efficacy, safety, and PK of an oral daily dosing regimen in both pediatric and adult patients with PROS.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Participants are assigned to age-based cohorts)

3.2 Planned Enrollment & Duration Target \$\$\$: Approximately 104 participants **Number of Sites:** Global, Multicenter **Estimated Study**

Start: October 2025 Estimated Primary Completion: Q4 2026
(Estimated duration for 48-week core period)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants aged ≥ 2 years at the time of informed consent/assent. 2. Diagnosis of PROS (according to Clinical Diagnostic Criteria for PROS proposed by Keppler Noreuil et al 2014) with symptomatic AND progressive overgrowth, who have syndromic disease or isolated features (with the exception of isolated macrodactyly, macrocephaly, or epidermal nevus). 3. Documented evidence of a somatic mutation(s) in the *PIK3CA* gene performed in local laboratories using a DNA-based test AND available archival tissue (if archival tissue is not available, a fresh biopsy should be performed if not clinically contraindicated). 4. Karnofsky (in participants >16 years of age) or Lansky (≤ 16 years of age) performance status index ≥ 50 . 5. Patient Global Impression of Severity (PGI-S) score of mild, moderate, severe, or very severe at screening. 6. Adequate bone marrow and organ function. 7. Presence of at least 1 PROS-related measurable lesion (longest diameter ≥ 2 cm) confirmed by BIRC assessment and associated with complaints, clinical symptoms, or functional limitations affecting everyday life.

4.2 Exclusion Criteria 1. Participants with only isolated macrodactyly, epidermal nevus/nevi, and macroencephaly (the only clinical feature or a combination of any of three) in the absence of other PROS-related lesions. 2. Previous treatment with alpelisib and/or any other phosphatidylinositol 3-kinase (PI3K) inhibitor (except treatment attempt defined as duration < 2 weeks and stopped at least 4 weeks prior to the first dose). 3. Debulking or other major surgery performed within 3 months of informed consent/assent. 4. Radiation exposure for PROS treatment purposes within 12 months prior to informed consent/assent. 5. Clinically meaningful PROS-related thrombotic event (Grade 2 and more as per CTCAE v4.03) within 30 days before informed consent/assent, and/or sclerotherapy/embolization for vascular complications performed within 6 weeks before informed consent/assent. 6. Clinically meaningful bleeding from a PROS-related lesion (Grade 2 and more as per CTCAE v4.03) within 30 days before study treatment initiation. 7. Participants with clinically significant worsening of PROS-related laboratory abnormalities, physical signs, and symptoms (such as, but not limited to, increase of D-dimers, worsening of underlying pain, newly occurring swelling or redness) indicating an uncontrolled condition during screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Group 1 (Adults ≥ 18 years):** Alpelisib 250 mg once daily (no dose escalation allowed). * **Group 2 (Children & Adolescents 2 to < 18 years):** Alpelisib 50 mg once daily for ages 2 to < 6 years; 125 mg once daily for ages 6 to < 18 years. **Route of Administration:** Oral
Duration of Treatment: * **Core Period:** 48 weeks. * **Extension 1 Period:** Up to Week 168 following the completion of the core period. * **Extension 2 Period:** Continues until the last participant completes Extension 2. Participants still deriving clinical benefit from alpelisib at the end of the study may receive Post-Trial Access (PTA).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures for PROS-related complications.
Prohibited: Concomitant use of other investigational therapies or PI3K inhibitors during the study duration.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -42	Informed Consent, Medical History, Genotyping, Whole Body MRI
Baseline	Day 0	Baseline MRI Assessment, First Dose of Alpelisib
Core Period	Up to Week 48	Daily dosing, Safety Labs, PK Sampling, Efficacy Assessments
Extension 1	Up to Week 168	Continued treatment under core rules for durable response assessment
Extension 2	Post Ext 1	Long-term continuation until study conclusion; potential transition to Post-Trial Access (PTA)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants with a confirmed objective response by BIRC. **Measurement Method:** Confirmed objective response requires achieving a radiological response, confirmed by a subsequent assessment performed at least after 4 weeks. Radiological response requires a $\geq 20\%$ reduction from baseline in the sum of target lesion volumes (1 to 3 target lesions) assessed by MRI BIRC, provided that none of the individual target lesions has a $\geq 20\%$ increase from nadir, in the absence of progression of non-target lesions, and without new lesions.

7.2 Secondary Endpoints * **Target Lesions:** Change from baseline (as assessed by BIRC) in target lesion volume. * **Non-Target Lesions:** Change from baseline (as assessed by BIRC) in MRI-measurable non-target lesion volume. * **All Lesions:** Change from baseline (as assessed by BIRC) in all MRI-measurable (target and non-target) lesion volume.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events, particularly focusing on known class effects of PI3K inhibitors such as hyperglycemia and rash. **Serious Adverse Events (SAEs):** Routine SAE reporting procedures. **Data Monitoring Committee (DMC):** Regular safety reviews conducted by an independent data monitoring committee to assess the risk-benefit profile across age cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy evaluating the confirmed objective response. Safety Set for all treated patients. **Sample Size Justification:** A sample size of $N=104$ is targeted to power the primary objective of detecting meaningful target lesion reductions. **Handling of Missing Data:** Standard handling mechanisms, including Last Observation Carried Forward (LOCF) for clinical response where protocol-defined.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Routine remote and onsite monitoring to ensure protocol adherence and accurate data collection. **Audit Trail:** Centralized BIRC verification for MRIs to ensure unbiased structural assessments; fully auditable eCRF system for PK and safety data.

11. Study Identifiers & Governance

Primary ID: CBYL719F12202 **Registry Number:** NCT06997588 **Sponsor/Lead Organization:** Novartis **Study Title:** EPIK-P4 **Official Regulatory Title:** Study Assessing the Efficacy, Safety and Pharmacokinetics of Alpelisib in Pediatric and Adult Patients With PIK3CA-related Overgrowth Spectrum (PROS)

12. Clinical Framework (PROS Specific)

Primary Condition: PIK3CA-related Overgrowth Spectrum (PROS) **Diagnosis Threshold:** Keppler Noreuil et al 2014 Criteria + Confirmed *PIK3CA* somatic mutation + Target lesion ≥ 2.0 cm **Medical Methodology:** PI3K α -specific small-molecule inhibitor targeting the AKT/mTOR pathway **Optimization Target:** Confirmed objective response ($\geq 20\%$ reduction) via BIRC MRI

13. Study Procedures & Methodology

Management Pathway: Stepwise evaluation using Whole Body MRI at baseline and defined intervals **Education Protocol (HCP):** BIRC imaging protocols and administration guidelines for weight/age-tiered dosing **Education Protocol (Patient):** Guidance on managing common adverse events associated with PI3K inhibitors **Quality Audit Mechanism:** Independent blinded centralized reading of MRI scans to limit investigator bias

14. Observation & Follow-Up Schedule

Total Duration: 48 weeks core + Extension 1 (up to Week 168) + Extension 2 (until study closure) **Onsite Visit Cadence:** Predefined assessment intervals across the 48-week core **Remote Monitoring Frequency:** Regular safety follow-ups **Checklist Review Interval:** Routine PK and safety laboratory monitoring

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				